

Muhammad Shayan Nazeer

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RESEARCH PROFILE

Ph.D. researcher in networks and distributed systems building low-latency infrastructure for mobile and edge AI. My work spans programmable 5G/6G core networks, distributed state consistency, fault-tolerant control planes, RAN mobility, KV-cache optimization, and test-time scaling for LLM serving at the mobile edge.

EDUCATION

PhD, Electrical and Computer Engineering

University of Massachusetts Amherst, USA

Sep. 2023 – Dec 2027 (Expected)

Advisor: [Dr. Taqi Raza](#)

- Research focus: networks and distributed systems for programmable mobile infrastructure, distributed state consistency, low-latency orchestration, and AI serving at the edge.

Bachelor's in Electrical Engineering

National University of Sciences and Technology (NUST), Islamabad, Pakistan

Sep. 2019 – Jul. 2023

Gold Medal

- Focus: wireless networks, embedded systems, machine learning, signal processing, computer architecture, and system security.

PATENTS

Method for Early Synchronization Through Source Cell for RACH-Less LTM

Muhammad Shayan Nazeer, A. Kak, N. Choi

Accepted, Nokia Patents Board

RESEARCH EXPERIENCE

University of Massachusetts Amherst

Graduate Research Assistant

Amherst, MA, USA

Sep. 2023 – Present

- **CoreX:** Built a distributed 5G/6G core that executes mobile-control procedures as DAG-scheduled workers over consistency-aware shared state; reduced registration latency by up to 2.1x, paging by 10.4x, and N2 handover by 16.6x, and improved failure-recovery p99 from 622 ms to 17 ms.
- **LLM Edge Serving:** Designing semantic KV-cache management for long-context LLM inference at the edge under GPU memory, network bandwidth, and latency constraints; techniques include topic-aware cache organization, precision-adaptive eviction, and cache merging to reduce memory pressure while preserving reasoning quality.
- **Test-Time Scaling:** Studying policies that dynamically allocate compute across prompts, models, and device–edge–cloud execution paths based on query difficulty, latency SLOs, and uncertainty estimates.

Nokia Bell Labs

Research Scientist Intern

Murray Hill, NJ, USA

Feb. 2025 – Jun. 2025

- Built a **reinforcement learning mobility framework** that leverages lower-layer radio measurements (MAC/RRC) to pre-synchronize target cells before handover execution, eliminating late synchronization in 5G NR mobility procedures.
- Implemented Lower Layer Triggered Mobility (LLT-M) end-to-end across 3 protocol layers—MAC, RRC, and control-plane signaling—validated over an experimental 5G NR testbed.
- Designed a RACH-less handover protocol that removes target-cell random-access signaling, reducing mobility overhead by 25% and cutting handover interruption time.
- Co-invented a Nokia Bell Labs patent on early synchronization and RACH-less mobility for 5G/6G RAN systems; accepted by Nokia Patents Board.

NUST NeuroInformatics Research Lab

Machine Learning Engineer

Islamabad, Pakistan

Sep. 2022 – Aug. 2023

- Built a real-time EEG inference pipeline for motor-intention decoding achieving sub-100 ms end-to-end latency across 4 stages: signal acquisition, preprocessing, deep-learning classification, and robotic actuation.
- Designed a hybrid CNN–RNN classifier that uses convolutional layers to extract spatial features across EEG channels and recurrent layers to model temporal dynamics across time steps, improving motor-intention discrimination over single-architecture baselines.
- Integrated the inference stack with an assistive robotic control system under hard real-time constraints; work published at IEEE IC2AI 2025.

PUBLICATIONS

- KV Cache Is the New User Context: Mobility-Aware LLM Serving at the Cellular Edge under review
Muhammad Shayan Nazeer, MT Raza
- CoreX: Next-G Cellular Core Network for Emerging Applications under review
Muhammad Shayan Nazeer, MT Raza
- FedAir: Network-Aware Federated Learning over 5G ACM MobiCom 2026, under review
MA Soomro, Muhammad Shayan Nazeer, FM Anwar, MT Raza
- Characterizing Cellular Network Dynamics for Federated Human Activity Recognition FmSys @ ACM SenSys 2026
MA Soomro, Muhammad Shayan Nazeer, MT Raza, FM Anwar
- SynchroNB: Toward Robust Timing for 5G NB-IoT Networks ACM SenSys 2026
MA Soomro, Muhammad Shayan Nazeer*, C. Delsignore, Y. Chandio, MT Raza, FM Anwar*
- ORACLE: Reconciling Next-G Cellular Core Operations for Correctness IEEE ICC 2026
Muhammad Shayan Nazeer, MT Raza
- Non-Invasive EEG-Driven Brain-Computer Interface for Motion Control of Assistive Robotics IEEE IC2AI 2025
S. E. Fatima, S. A. Rahman, W. Mumtaz, Muhammad Shayan Nazeer

TECHNICAL SKILLS

- **Programming:** Python, C, C++, Bash, JavaScript, MATLAB
- **Systems:** Distributed Systems, Fault Tolerance, State Consistency, Scheduling, Linux, Docker, Kubernetes, Git
- **Networking:** 5G/6G Core Networks, RAN Mobility, Control-Plane Design, SDN, NFV, Traffic Engineering, Protocol Analysis
- **AI Infrastructure:** LLM Serving, KV-Cache Optimization, Test-Time Scaling, Edge Inference, Model Serving, Device-Edge-Cloud Offloading
- **Machine Learning:** PyTorch, TensorFlow, Hugging Face Transformers, Reinforcement Learning, Federated Learning, RAG, Pandas, NumPy

HONORS & AWARDS

- NSF CONEXT Travel Grant** 2025
Awarded travel support to attend ACM CoNEXT.
- Colosseum Young Gladiator** 2024
NSF-funded master class on the Colosseum 5G testbed, Northeastern University.
- Rector's Gold Medal** 2023
Awarded for best undergraduate senior design project, NUST.
- NUST High Achiever's Award** 2023
Gold Medal for outstanding academic performance.
- CSAW Logic Locking Conquest** 2022
Global Runner Up, organized by NYU.
- National Physics Talent Contest** 2018
Selected among top 50 students in Pakistan for IPhO training.
- Intel ISEF** 2017
National-level finalist.

REFERENCES

- Dr. Muhammad Taqi Raza** — Ph.D. Advisor
Assistant Professor, ECE, University of Massachusetts Amherst
taqi@umass.edu
- Dr. Nak Jung Choi** — Research Internship Supervisor
Department Head, Mobile Network Systems, Nokia Bell Labs, Murray Hill, NJ
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- Dr. Fatima M. Anwar** — Research Collaborator
Associate Professor, ECE, University of Massachusetts Amherst
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